

BHUVAN GOWDA

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PROFILE

Software Engineer specializing in scalable full-stack systems and AI-powered applications, with strong expertise in Python, FastAPI, and React. Experienced in designing microservices and cloud-native architectures on AWS, with hands-on work in containerized deployments using Docker and Kubernetes. Currently pursuing an MSc in Artificial Intelligence at Technological University of the Shannon, Ireland.

KEY SKILLS

- **Programming Languages:** Python, Java, TypeScript/JavaScript, SQL
- **Web & Backend:** FastAPI, Django, Node.js, Next.js, Express.js, React, HTML, CSS
- **Databases:** MySQL, MongoDB, PostgreSQL, DynamoDB
- **Cloud Platforms:** AWS (SAA-C03), GCP, Azure, Terraform
- **AI / LLM Development:** LangChain, LangGraph, Hugging Face, ReAct Agents, OpenAI API, Vector DBs, RAG
- **Other Tools:** Git, GitHub, Docker, Kubernetes, Prometheus, Grafana, Loki, CI/CD, PySpark, Linux (RHEL), Pytest

WORK EXPERIENCE

Software Engineer Intern | TUS-Research Institute, Ireland

May 2026 - Present

- Built a serverless AWS pipeline (Lambda, S3, Python, pydicom) to automate DICOM ingestion and MRI quality assurance, eliminating 100% of manual QA effort and enabling audit-ready reporting.
- Developed a Python/MATLAB pipeline computing SNR, geometric accuracy, and signal uniformity metrics, integrated with AWS S3 for centralised DICOM metadata storage and efficient cross-system querying.

System Software Engineer - Intern | Red Hat, India

Jan 2025 - Jun 2025

- Automated RHEL infrastructure provisioning (user/group management, LVM, filesystem allocation) using Ansible and Bash, improving system reliability by 30% across production environments.
- Hardened enterprise security posture by implementing SELinux policies, ACLs, and firewall rules aligned with compliance standards, while contributing to sprint delivery within an Agile team.

Backend Developer - Intern | Maple labs, India

Jan 2025 - Jun 2025

- Developed and deployed RESTful APIs using FastAPI and Python, leveraging AWS EC2, Lambda, and S3 to deliver scalable cloud-based services and accelerate backend feature development.
- Enhanced application reliability and performance by testing, debugging, and optimizing backend services on AWS, improving system stability, reducing service disruptions, and enabling seamless integration of new features.

PROJECTS

SafeKube: Policy-constrained LLM Based Closed Loop Remediation in Kubernetes Clusters | [GitHub](#)

- Reduced MTTR by 15–150x by building SafeKube, an LLM-powered autonomous Kubernetes remediation agent integrated with Prometheus, Loki, and Grafana, achieving 100% fault resolution across 38 fault injection scenarios via Chaos Mesh.
- Improved remediation reliability and auditability by designing a multi-layer Policy Engine and Risk Scorer with human escalation pathways, automating fault detection across 7 fault categories through a closed-loop agentic decision system.

SQL-Copilot: AI Powered SQL Agent | [GitHub](#)

- Reduced manual query writing time by 70% by architecting an AI-powered SQL Copilot using FastAPI and Google Gemini 2.5 Flash, enabling natural language-to-SQL conversion via LangChain ReAct Agents with sub-2-second query execution.
- Eliminated manual database mapping by implementing SQLDatabaseToolkit for automated schema discovery, reducing query setup overhead by 60% across multi-table workflows while maintaining multi-turn context accuracy via ConversationBufferMemory.

EDUCATION

- MSc in Software Design with AI | TUS Athlone | Sept 2025 - Sept 2026 | First Class Honours (1:1)
- BE in Computer Science | Acharya Institute of Technology | Dec 2021 - Jun 2025 | CGPA: 9.2/10

CERTIFICATIONS

- AWS Certified Solutions Architect - Associate (2026)
- RedHat Certified System Administrator (2025)
- Oracle Certified AI foundations Associate (2025)